SOST20131 Assessment Part II

Words count 976 + 1012 = 1988

1SOST20131 Assessment Part II *1*

2 Question 1: Multiple Regression *2*

2.1 Research questions **2**

2.2 Data Description and Exploratory Analysis **2**

2.3 Initial model specification **5**

2.4 Model selection and refinement **7**

2.5 Final model presentation and interpretation **10**

2.6 Prediction and Comments on Quality **12**

2.7 Brief Conclusion **12**

3 Question 2: Logistic Regression *13*

3.1 Research question: **13**

3.2 Data Description and Exploration **13**

3.3 Initial Logistic Regression Model **18**

3.4 Model selection and Goodness of fit **20**

3.5 Final Logistic Model and Interpretation **23**

3.6 Classification Accuracy and Model Performance **25**

3.7 Limitations and brief conclusion **25**

4. Appendix **26**

2 Question 1: Multiple Regression

2.1 Research questions

The main research question is to examine the relationship between Sales preference scores and the other relevant variables.

Specifically, the research will attempt to understand the following question:

What are the statistically significant effects of background factors and ability test results on sales performance?

Statistically, the response variable is Sales_Perf, denoted as Y, and the explanatory variables, shown as X's, where e represents the error term($e \sim N(0,\sigma 2)$).

2.2 Data Description and Exploratory Analysis

The dataset selects 48 employees randomly, types of variables are summarised in Table 1.

Explanatory variable (X's)	Types of variables
Creativity test score (Creativity, X1)	Measured
Logical reasoning test score (Mechanical, X_2)	Measured
English test score (Abstract, X3)	Measured
Numeracy test score (Maths, X_4)	Measured
Background type coded 0 = Science, 1 = Business (Type, X_5)	Categorical
Recording the level of education 1= Degree, 2 = Masters, 3 = Doctorate, (Level, X_6)	Categorical
Recording the employee's main functional area 1 Logistics, 2 Promotion, 3Sales, 4 Strategy, (Expertise, X_7)	Categorical

Table 1

Note: The categorical variable will interpret the possible effects by comparing to the baseline group.

To gain an overview of the data, Table 2 shows basic summary statistics for the response variable and measured variables.

Variable Type	**Variable Name**	**Mean**	**Std.Dev**	**Min**	**Max**
Response	Sales_Perf	106.3	9.85	89.8	122.3
Measured	Creativity	10.98	3.91	1.00	18.00
Measured	Mechanical	14.21	3.53	5.00	20.00
Measured	Abstract	11.35	1.47	8.00	15.00
Measured	Maths	29.12	10.50	9.00	51.00

Table 2

Note: Measured in 2 d.p.

Pairwise relationships and distributions are illustrated using a ggpairs plot (**Figure 1**). Based on Figure 1, the Sales_pref is approximately symmetric without any obvious skewness. For explanatory variables, on one hand, the distributions of Creativity, Mechanical and Maths demonstrate around the centre, despite the images of Mechanical and Maths showing a modest right-skew and double peaks. Conversely, the Abstract shows multiple elevations and numerous "up and down" waves, indicating volatility in the relationship.

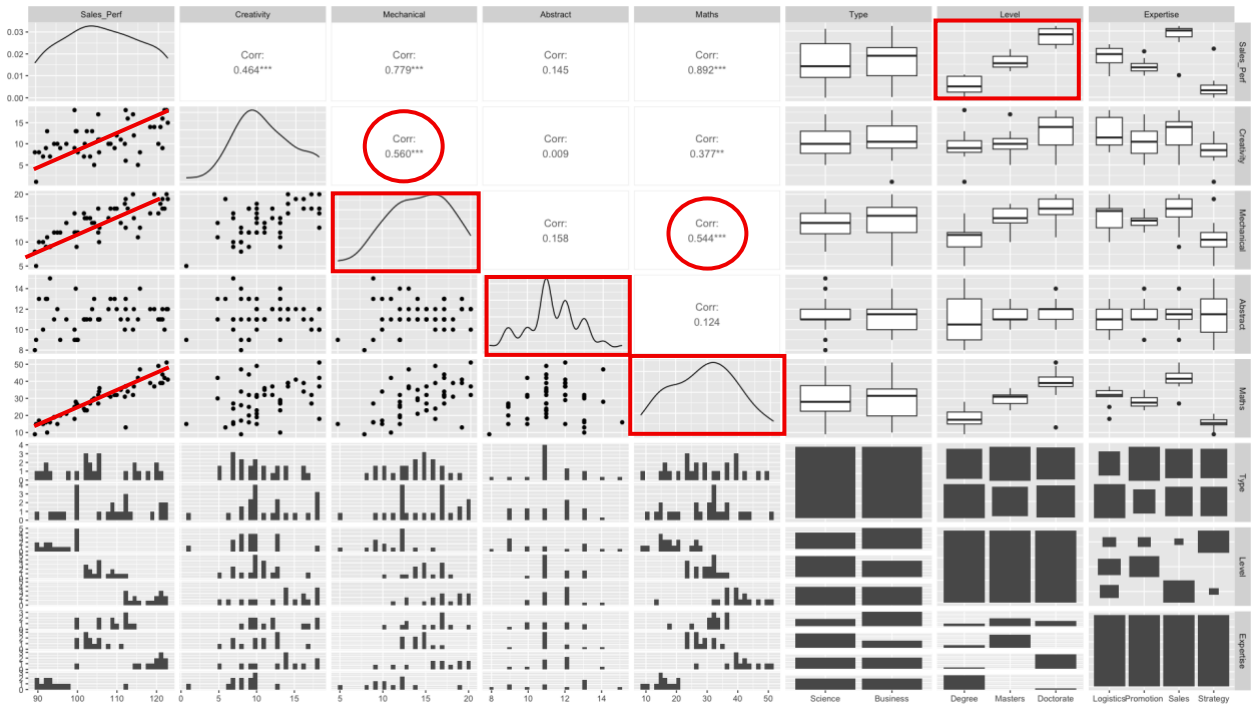

Figure 1

Figure 2 is a side-by-side boxplots of Sales_Perf by Type, Level, and Expertise, which are all categorical variables (A clear zoom version). These plots indicate systematic differences across groups, by education level and Expertise area.

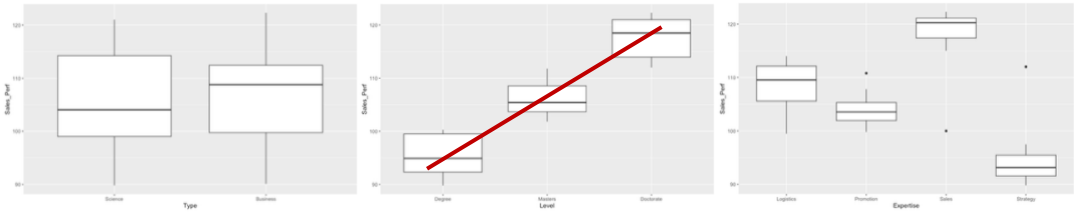

Figure 2

Explanatory variable	Prior knowledge/viewpoint	Sample evidence (strength)
Creativity (Creativity, X1)	Positive +	Positive + (moderate)
Logical reasoning test (Mechanical, X2)	Positive +	Positive + (strong)
English test (Abstract, X_3)	Positive +	Weak
Numeracy test (Maths, X_4)	Positive +	Positive + (strong)
Background type (Type, X_5)	Business > Science	Difference by group.
Education level (Level, X_6)	Unclear	Weak
Primary function area (Expertise, X_7)	Rank: 1 Sales 2 Promotion 3 Strategy 4 Logistics	Difference by group.

Table 3

A summary of prior expectations and sample evidence is provided in **Table 3**. Overall, most of the findings match our social intuition. Sales performance is strongly correlated with higher logical reasoning and numeracy. But the significant correlations between Mechanical and Maths (Corr ≈ 0.54) and Creativity and Mechanical (Corr ≈ 0.56), inspire the use of vif/gvif for further checking in section 2.3 for multicollinearity.

Taken together, the exploratory study confirms that a multiple linear regression approach is appropriate.

2.3 Initial model specification

The initial model includes all the variables in our model, avoiding potential confounding effects and omitted variable bias. This also allows for testing joint significance level.

Initial model:

$$\textbf{Sales_Perf} = \boldsymbol{\beta_0} + \boldsymbol{\beta_1}\,\textbf{Creativity} + \boldsymbol{\beta_2}\,\textbf{Mechanical} + \boldsymbol{\beta_3}\,\textbf{Abstract} + \boldsymbol{\beta_4}\,\textbf{Maths} + \boldsymbol{\beta_5}\,\textbf{Type} + \boldsymbol{\beta_6}\,\textbf{Level} + \boldsymbol{\beta_7}\,\textbf{Expertise} + \boldsymbol{e}$$

Measure variables are plugged into model directly, while categorical variables are included using clearly specified reference groupings and dummy coding.

For the Sales_Perf regression model versus Type, to improve our visuality, I will simplify the model using $\beta_n X_n$ to represent all the rest variables* their coefficient,

$$\text{Sales_Perf} = \beta_0 + \sum_n \beta_n X_n + \beta_5\,\text{Type} + e$$

$$\text{Sales_Perf} = \beta_0 + \textstyle\sum_n \beta_n X_n + \beta_5 * 1 + \text{e} = \beta_0 + \textstyle\sum_n \beta_n X_n + \beta_5 + \text{e},$$

$for\ employee\ has\ a$ **Business** $background$

$$\text{Sales_Perf} = \beta_0 + \sum_n \beta_n X_n + \beta_5 * 0 + \text{e} = \beta_0 + \sum_n \beta_n X_n + \text{e}$$

$for\ employee\ has\ a$ **Science** $background$

The value of β_5 represents the average difference in Sales preference scores between **Business** $background\ and$ **Science** $background.$

Degree is served as the reference category for Level (X6), and the dummy coefficients measure mean differences in Sales_Perf relative to Degree holders; a similar interpretation to Expertise (X7).

reference group: Level = 1 → Degree
Level2 = 1 → Master, otherwise 0
Level3 = 1 → Doctorate, otherwise 0
reference group: Expertise = 1 → Logistics
Expertise 2 = 1 → Promotion vs Logistics
Expertise 3 = 1 → Sales vs Logistics
Expertise 4 = 1 → Strategy vs Logistics

In general, this model fits the data well (R^2 = 0.9806, Adjusted R^2 = 0.9753), which means 98.1% of the changes in Y are explained by changes in X's, and it is jointly statistically significant (F(10, 37) = 186.7, p < 0.001).

The intercept creates the average score for a "baseline employee with a Science background, a Degree qualification, and a Logistics role, would perform a score around 83.66, providing a basic understanding of our model.

Mechanical ($\beta = 0.54$) and Maths ($\beta = 0.27$) are both positively and highly significantly associated with Sales_Perf ($p < 0.001$).

In contrast, Creativity ($p \approx 0.41$) and Abstract ($p \approx 0.84$) are not statistically significant ($p > 0.1$).

Regarding dummy variables, Business background is associated with a 1.62-point higher Sales_Perf than Science (SE ≈ 0.49, $p \approx 0.002$). Relative to Degree holders, Master's and Doctoral holders increase Sales_Perf by approximately 4.68 and 10.03 points, respectively (both $p < 0.001$).

For Expertise, only employees who work in Sales ($\hat{b} \approx 3.08$ SE ≈ 0.90 $p \approx 0.00162$) are more efficient higher than those in Logistics.

2.4 Model selection and refinement

Before simplifying the model, multicollinearity was assessed using vif. As a strong correlation has been tested in 2.2.

```
> # check vif to avoid multicollinearity
> library(car)
Loading required package: carData

Attaching package: 'car'

The following object is masked from 'package:dplyr':

    recode> vif(model_initial)
                GVIF Df GVIF^(1/(2*Df))
Creativity  1.877386  1        1.370177
Mechanical  3.109157  1        1.763280
Abstract    1.169796  1        1.081571
Maths       7.070664  1        2.659072
Type        1.187312  1        1.089638
Level      12.560609  2        1.882577
Expertise  16.255324  3        1.591595
```

Hence, there is no strong multicollinearity in our initial model, supported by checking the GVIF^(1/(2*Df)), almost no bigger than 2. Thus, multicollinearity is not a concern and does not affect the subsequent process.

Model refinement was conducted by sequentially deleting statistically insignificant variables. Abstract was removed (model_1) and a slight improvement in the R^2(0.9805), adjusted R^2(0.9759), endorsing its exclusion.

```
> # Delete Abstract
> model_1 <- update(model_initial, . ~ . - Abstract)
> summary(model_1)

Call:
lm(formula = Sales_Perf ~ Creativity + Mechanical + Maths + Type +
    Level + Expertise, data = emp)

Residuals:
    Min      1Q  Median      3Q     Max
-2.7198 -0.7457 -0.0734  0.9616  3.3653

Coefficients:
                   Estimate Std. Error t value Pr(>|t|)
(Intercept)        83.41804    2.18987  38.093  < 2e-16 ***
Creativity         -0.06585    0.07817  -0.842  0.40486
Mechanical          0.69252    0.11092   6.244 2.64e-07 ***
Maths               0.26754    0.05533   4.835 2.22e-05 ***
TypeBusiness        1.62037    0.47994   3.376  0.00171 **
LevelMasters        4.65938    1.00708   4.627 4.23e-05 ***
LevelDoctorate     10.00660    1.16051   8.623 1.77e-10 ***
ExpertisePromotion -0.59802    0.71768  -0.833  0.40991
ExpertiseSales      3.09324    0.88953   3.477  0.00128 **
ExpertiseStrategy  -1.51461    1.10955  -1.365  0.18026
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.528 on 38 degrees of freedom
Multiple R-squared:  0.9805,    Adjusted R-squared:  0.9759
F-statistic: 212.8 on 9 and 38 DF,  p-value: < 2.2e-16
```

The second least significant variable is Creativity, which will be deleted in (model_2). From the result, model_2 is a more parsimonious model (R^2(0.9802), adjusted R^2(0.9761)).

```
> # Delete Creativity
> model_2 <- update(model_1, . ~ . - Creativity)
> summary(model_2)

Call:
lm(formula = Sales_Perf ~ Mechanical + Maths + Type + Level +
    Expertise, data = emp)

Residuals:
    Min      1Q  Median      3Q     Max
-2.6409 -0.8593  0.0555  0.9615  3.2718

Coefficients:
                   Estimate Std. Error t value Pr(>|t|)
```

```
(Intercept)        83.02821    2.13243  38.936  < 2e-16 ***
Mechanical          0.65100    0.09899   6.576 8.22e-08 ***
Maths               0.26784    0.05513   4.859 1.95e-05 ***
TypeBusiness        1.60600    0.47785   3.361 0.001749 **
LevelMasters        4.97227    0.93259   5.332 4.38e-06 ***
LevelDoctorate     10.15796    1.14223   8.893 6.36e-11 ***
ExpertisePromotion -0.56683    0.71405  -0.794 0.432107
ExpertiseSales      3.16870    0.88171   3.594 0.000903 ***
ExpertiseStrategy  -1.21939    1.04882  -1.163 0.252043
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.522 on 39 degrees of freedom
Multiple R-squared:  0.9802,	Adjusted R-squared:  0.9761
F-statistic: 241.1 on 8 and 39 DF,  p-value: < 2.2e-16
```

An alternative specification deleting Expertise substantially reduced model fit, indicating that this variable should be retained.

```
> # Delete Expertise
> model_3 <- update(model_1, . ~ . - Expertise )
> summary(model_3)

Call:
lm(formula = Sales_Perf ~ Creativity + Mechanical + Maths + Type +
    Level, data = emp)

Residuals:
    Min      1Q  Median      3Q     Max
-3.7455 -0.7464 -0.0415  1.2256  2.9772

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)    79.64811    1.46264  54.455  < 2e-16 ***
Creativity     -0.06371    0.08277  -0.770 0.445910
Mechanical      0.73083    0.12433   5.878 6.45e-07 ***
Maths           0.39965    0.04140   9.654 4.08e-12 ***
TypeBusiness    1.54975    0.52782   2.936 0.005428 **
LevelMasters    3.54556    0.96886   3.660 0.000714 ***
LevelDoctorate 10.02529    1.29794   7.724 1.60e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.722 on 41 degrees of freedom
Multiple R-squared:  0.9734,	Adjusted R-squared:  0.9695
F-statistic: 249.7 on 6 and 41 DF,  p-value: < 2.2e-16
```

In conclusion, Model 2 offers an ideal model, due to its strong interpretability, extremely high explanatory power, substantial statistical significance for important variables, and high adjusted, theoretically reasonable coefficients.

2.5 Final model presentation and interpretation

Based on the following result, here is our final model:

$$\widehat{Sales_Perf} = 83.0282 + 0.6510\,Mechanical + 0.2678\,Maths + 1.6060\,Busines + 4.9723\,Masters + 10.1580\,Doctorate - 0.5668\,Promotion + 3.1687\,Sales - 1.2194\,Strategy$$

The final model explains approximately **98%** of the variation in sales performance and remains jointly significant.

Holding other variables constant,

Continuous predictor:

Mechanical (β = 0.65) and Maths (β = 0.27), higher Mechanical score or Maths score is associated with better performance on average.

Type:

Employees with a Business background score, on average, earn 1.61 points higher than Science employees, controlling for other variables.

Level:

Compared with Degree holders, master's graduates score about 5 points higher and Doctorate graduates score about 10 points higher, on average.

Expertise:

Relative to variable Logistics, employees in Sales perform around 3.17 points higher, while Promotion and Strategy roles do not demonstrate a significant difference. But the Expertise are significant as a “whole” variable.

The residual plots **(Figure 3)** show an approximately central distribution, supported by the standard error of 1.52, and no serious outliers.

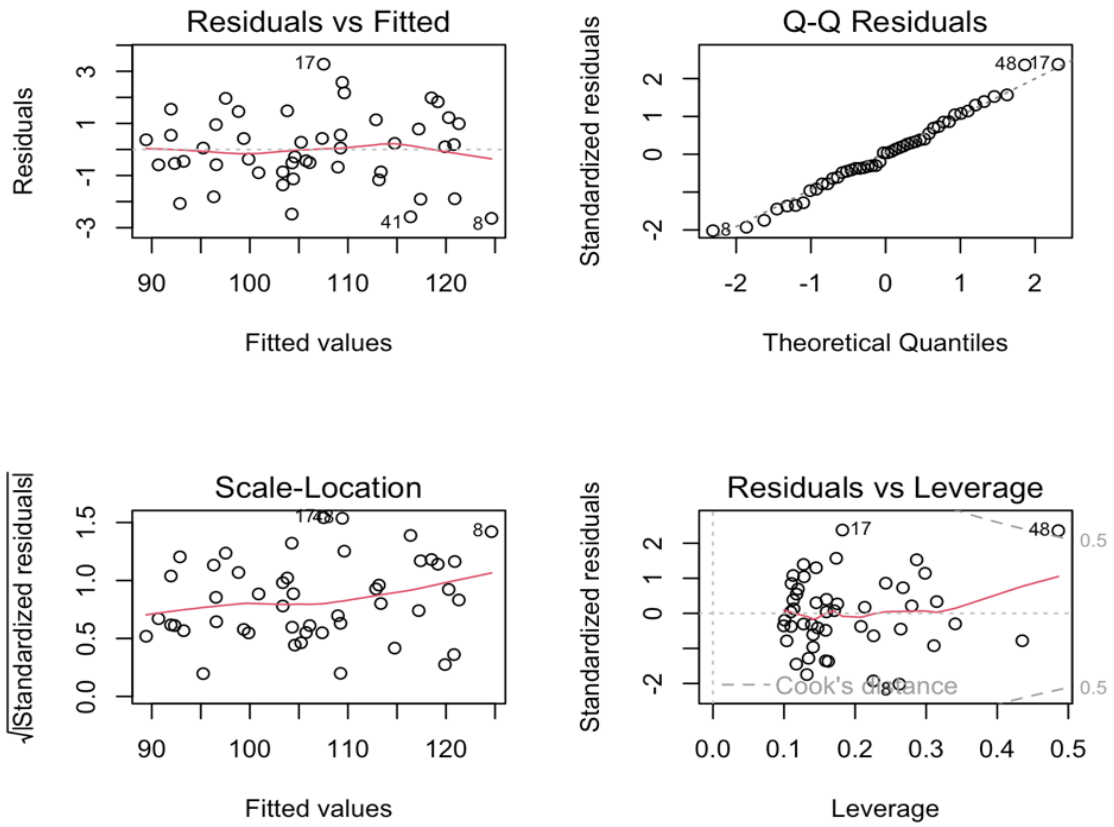

Figure 3

2.6 Prediction and Comments on Quality

Sales_Perf = 83.0281 + 0.6510 * 18 + 0.2678 * 60 + 10.1580 -0.5668 = 120.41

Based on our final model, we can predict that the employee's performance is 120.41.

```
> Sales_Perf = 83.0281 + 0.6510 * 18 + 0.2678 * 60 + 10.1579 -0.5668

> Sales_Perf

[1] 120.4053
```

Excellent model fit is shown by an adjusted R^2 of 0.98, and the forecast is credible because the employee profile is within the sample data's support, which is typical.

Nevertheless, the relatively small sample size (n = 48) and the presence of residual variability(residual sd = 1.52) imply some uncertainty, these may lead to a small bias in our final prediction.

2.7 Brief Conclusion

In conclusion, further analyses are still necessary, especially when applying to social research questions.

3 Question 2: Logistic Regression

3.1 Research question:

The primary research question of this study is whether a set of health and lifestyle indicators is associated with the probability that an individual has prediabetes or diabetes or not.

Specifically, we examine how high blood pressure, high cholesterol, body mass index, smoking status, fruit consumption, and age relate to the probability $\mathrm{P}(\mathrm{Diabetes} = 1)$. Considering the response variable is binary, a logistic regression framework is implemented.

3.2 Data Description and Exploration

There are 150 observations in our dataset, two types of variables in this research, as shown in **Table 1**. The binary indicators shown on the left show the tester's healthy status, summarised by proportions and barplots. The measured variables are numeracy, suitable for summary statistics like mean, standard deviation, and boxplots to compare the difference between diabetes status.

Binary indicators

High BP	0 - no high blood pressure 1 - high blood pressure
HighChol	0 - no high cholesterol 1 - high cholesterol
Smoker	0 - no smoker 1 - smoker
Fruits	Consume fruit 1 or more times per day: 0 - no 1 – yes

Table 1

Measured variables

BMI	*Body Mass Index*
Age	*Age of the patient*

```
> # Function to create count & proportion tables
> binary_vars <- c("HighBP", "HighChol", "Smoker", "Fruits")
> lapply(binary_vars, function(var) {
+   diabetes %>%
+     group_by(Diabetes, .data[[var]]) %>%
+     summarise(n = n(), .groups = "drop") %>%
+     group_by(Diabetes) %>%
+     mutate(prop = n / sum(n))
+ })
Table[[2.1]]
# A tibble: 4 × 4
# Groups:   Diabetes [2]
  Diabetes HighBP     n  prop
     <int>  <int> <int> <dbl>
1        0      0    38 0.514
2        0      1    36 0.486
3        1      0    22 0.289
4        1      1    54 0.711
```

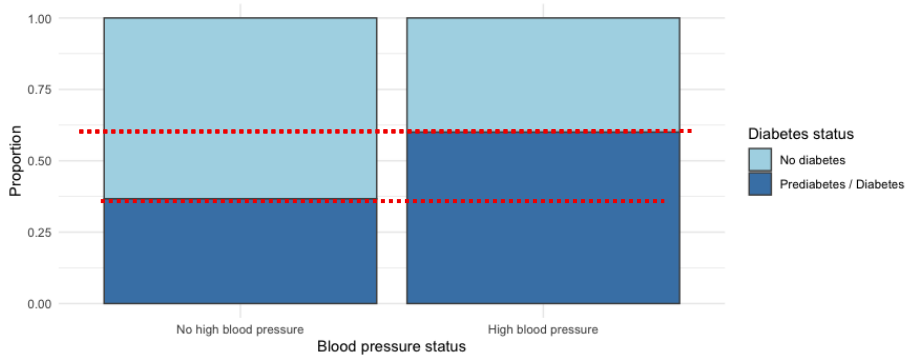

Figure 2.1

```
Table[[2.2]]
# A tibble: 4 × 4
# Groups:   Diabetes [2]
  Diabetes HighChol     n  prop
     <int>    <int> <int> <dbl>
1        0        0    44 0.595
2        0        1    30 0.405
3        1        0    24 0.316
4        1        1    52 0.684
```

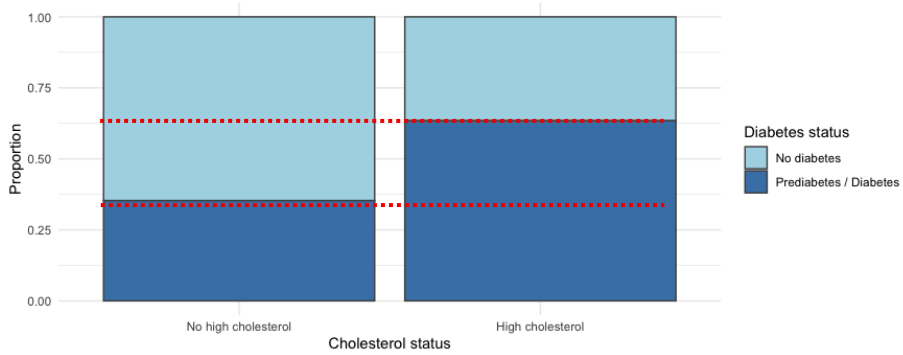

Figure 2.2

```
Table[[2.3]]
# A tibble: 4 × 4
# Groups:   Diabetes [2]
  Diabetes Smoker     n  prop
     <int>  <int> <int> <dbl>
1        0      0    43 0.581
2        0      1    31 0.419
3        1      0    38 0.5
4        1      1    38 0.5
```

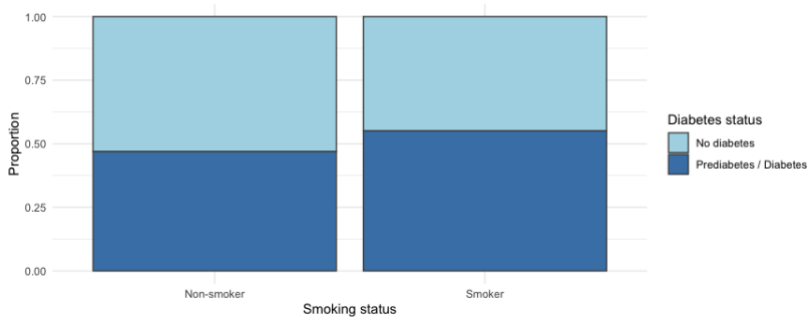

Figure 2.3

```
Table[[2.4]]
# A tibble: 4 × 4
# Groups:   Diabetes [2]
  Diabetes Fruits     n  prop
     <int>  <int> <int> <dbl>
1        0      0    30 0.405
2        0      1    44 0.595
3        1      0    29 0.382
4        1      1    47 0.618
```

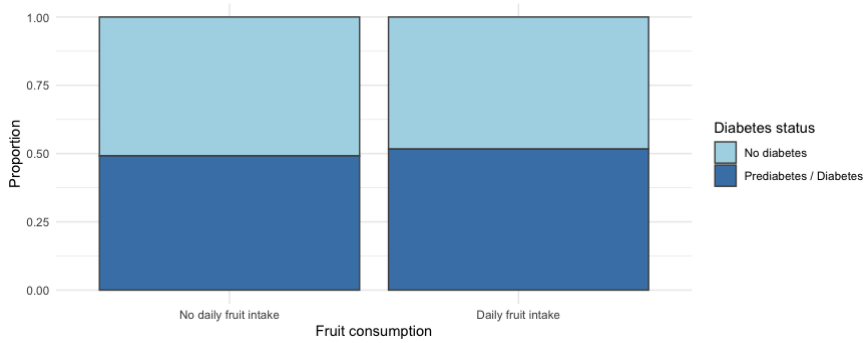

Figure 2.4

```
# Summary statistics for BMI and Age by diabetes status
# A tibble: 2 × 9
  Diabetes mean_BMI sd_BMI min_BMI max_BMI mean_Age sd_Age min_Age max_Age
     <int>    <dbl>  <dbl>   <int>   <int>    <dbl>  <dbl>   <int>   <int>
1        0     26.8   4.74      17      42     53.2   9.57      33      65
2        1     32.9   8.25      20      57     57.6   10.3      27      87
```

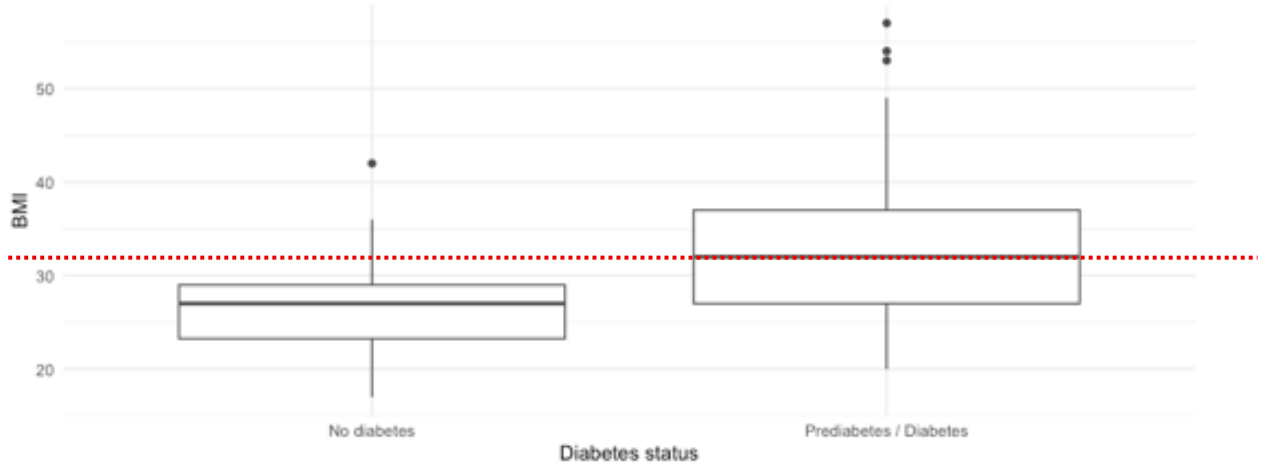

Figure 2.5

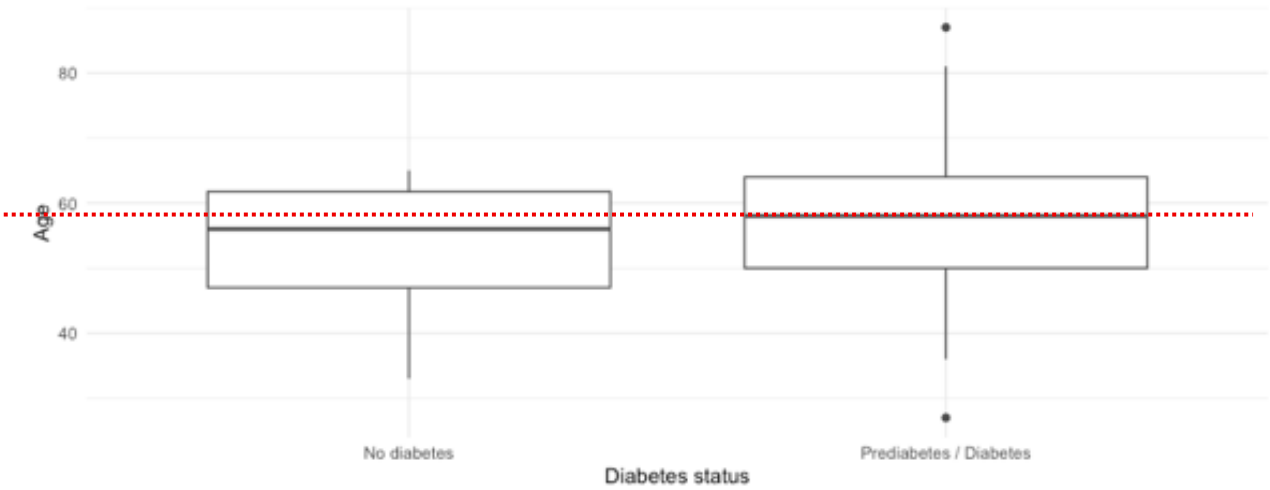

Figure 2.6

On balance, the variables HighBP, HighChol, and BMI may be related to higher diabetes risk.

Explanatory variable	Signs in the model	Discussion	Estimated Sign
High BP	Strong Positive +	Among the people with higher blood pressure, prediabetes accounted for higher proportion, around 71.1%. (Table [[2.1]], Figure 2.1)	$\beta_1 HighBP > 0$
HighChol	Strong Positive +	Individuals with high cholesterol, the proportion of people with diabetes is significantly higher approximately 68.4%. (Table [[2.2]], Figure 2.2)	$\beta_2 HighChol > 0$
Smoker	Weak	The diabetes ratio of people who smoked or not is similar(gap:8.1%). (Table [[2.3]], Figure 2.3).	$\beta_4 Smoker \geq 0$
Fruits	Weak	The difference (2.3%) of diabetes ratio between people who consumed fruit or not is quite small. (Table [[2.4]], Figure 2.4).	$\beta_5 Fruits \leq 0$
BMI	Extreme strong positive +	Both the mean (32.9 VS 26.8) and median (around 32) of BMI (Diabetes) are significantly higher than No diabetes (Figure 2.5, Table 2.5). The distribution of BMI(Diabetes) are right skewed, also proved by some extreme larger outliers.	$\beta_3 BMI > 0$
Age	Moderate	The individual who got diabetes' age is higher (57.6 > 53.2) and shows a positive trend. (Figure 2.6)	$\beta_6 Age > 0$

3.3 Initial Logistic Regression Model

The initial model is fitted with all six explanatory variables:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 HighBP + \beta_2 HighChol + \beta_3 BMI + \beta_4 Smoker + \beta_5 Fruits + \beta_6 Age$$

Before the model fitting procedure, the data is split into a proportion of training (80%) and testing (20%), for later assessing model accuracy in correct prediction using the test data set.

```
Check the balance
> prop.table(table(diabetes_test$Diabetes))
        0         1
0.3666667 0.6333333

> set.seed(123)
> split_idx = sample(nrow(diabetes), 120 )
>
> diabetes_train <- diabetes[split_idx, ]
> diabetes_test  <- diabetes[-split_idx, ]
```

The model is estimated using the glm() function, with the binomial and logit link functions.

Based on the result, BMI and Age are the variables that have a significant effect, with their p-values < 0.01 and 0.022, respectively. While HighChol is approximately marginally significant at the 10% level (P-value = 0.1060). In contrast, the variables HighBP, Smoker, and Fruits are all shown to be insignificant (p-values are far larger than 0.05). The Model fit improves compared to the null model, with deviance reduction from 166.06 to 130.29 and a corresponding AIC of 144.29.

The effects of BMI and Age are consistent with prior expectations: higher BMI and older age are both associated with increased diabetes risk. Combined with the ANOVA results, these variables are retained in the model.

Given the binary answer and the nonlinear relationship between Y and X, logistic regression is appropriate in this situation since it models the log-odds of the binary result as a linear function of the explanatory factors. In contrast, OLS is inappropriate due to violations of distributional assumptions. Moreover, no influential outliers are detected, supporting the validity of the model.

```
> model_initial <- glm(
+   Diabetes ~ HighBP + HighChol + BMI + Smoker + Fruits +
Age,
+   data = diabetes_train,
+   family = binomial(link = "logit")
+ )
> summary(model_initial)

Call:
glm(formula = Diabetes ~ HighBP + HighChol + BMI + Smoker +
Fruits +
    Age, family = binomial(link = "logit"), data =
diabetes_train)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -8.47648    1.97187  -4.299 1.72e-05 ***
HighBP       0.15824    0.47409   0.334 0.738556
HighChol     0.75810    0.46900   1.616 0.106003
BMI          0.15244    0.04171   3.655 0.000257 ***
Smoker       0.20972    0.42914   0.489 0.625058
Fruits       0.48404    0.44635   1.084 0.278170
Age          0.05472    0.02388   2.292 0.021915 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 166.06  on 119  degrees of freedom
Residual deviance: 130.29  on 113  degrees of freedom
AIC: 144.29

Number of Fisher Scoring iterations: 4
```

3.4 Model selection and Goodness of fit

To assess whether the explanatory variables collectively contribute to the model, a G statistic (likelihood ratio test) is conducted by comparing the null deviance with the residual deviance. Comparing this G statistic to the chi-squared distribution to test,

$$H_0\colon \beta_i = 0$$

$$H_1\colon \text{at least one is different from 0}$$

$$\text{where } i = 1,2,3,4,5,6.$$

$$\text{If } G_{calc} < G_{crit} \Rightarrow H_0, \text{ o.w. } H_1$$

As $Gcalc = 35.76544 > Gcrit = 12.5916 \Rightarrow H1$, ie, p-value < 0.001, indicating that this is a statistically valid model and that the variables collectively have explanatory power. Pseudo- measures (McFadden ≈ 0.22) further suggest reasonable explanatory power for a logistic model.

```
<- model_initial$null.deviance - model_initial$deviance
> Gdf    <- model_initial$df.null - model_initial$df.residual
>
> G_calc
[1] 35.76544
> Gdf
[1] 6
> G_crit <- qchisq(0.95, df = Gdf)
> G_crit
[1] 12.59159
>
> # 3) p-value
> p_value_G <- 1 - pchisq(G_calc, df = Gdf)
> p_value_G
[1] 3.061601e-06
```

The ANOVA chi-square test is used as a complementary tool for nested model comparison, but it depends on variable ordering.

```
> anova(model_initial, test = "Chisq")
Analysis of Deviance Table

Model: binomial, link: logit

Response: Diabetes

Terms added sequentially (first to last)


         Df Deviance Resid. Df Resid. Dev  Pr(>Chi)
NULL                       119     166.06
HighBP    1   5.5102       118     160.54   0.01891 *
```

```
HighChol   1   5.0530       117     155.49   0.02458 *
BMI        1  18.2351       116     137.26 1.952e-05 ***
Smoker     1   0.2568       115     137.00   0.61235
Fruits     1   0.9106       114     136.09   0.33995
Age        1   5.7997       113     130.29   0.01603 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Combining the two test results, according to the Wald z and p-values, (($if\,(|z| < 2)\ and\ p - value\ >\ 0.05$), insignificant variables will be removed, and it can be confirmed that Smoker and Fruits are two predictors with no significant effect. However, for HighBP and HighChol, we will delete them to give it a try and check their significance. The Variable selection proceeds sequentially, removing Smoker, Fruits (in red) and HighBP, HighChol (in blue).

Model fit is compared using the Akaike Information Criterion (AIC), where smaller values indicate a better fit. The AIC shows a monotonic decline, dropping from 142.5288 to 139.9792, indicating a small improvement in model fit. However, removing HighChol made our model worse; AIC increased again. Thus, HighChol is retained. Furthermore, deleting these variables can also be confirmed by their p-values, which are all extremely large, indicating that they are not significant variables.

All in all, the final model is HighChol + BMI + Age.

```
1. Remove Smoker

> anova(model_initial, m1, test = "LRT")
Analysis of Deviance Table

Model 1: Diabetes ~ HighBP + HighChol + BMI + Smoker + Fruits + Age
Model 2: Diabetes ~ HighBP + HighChol + BMI + Fruits + Age
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1       113     130.29
2       114     130.53 -1 -0.23909   0.6249
> AIC(model_initial, m1)
              df      AIC
model_initial  7 144.2898
m1             6 142.5288
```

```
2. Remove fruit
> anova(m1, m2, test = "LRT")
Analysis of Deviance Table

Model 1: Diabetes ~ HighBP + HighChol + BMI + Fruits + Age
Model 2: Diabetes ~ HighBP + HighChol + BMI + Age
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1       114     130.53
2       115     131.76 -1  -1.2337   0.2667
> AIC(m1, m2)
   df      AIC
m1  6 142.5288
m2  5 141.7625


3. Remove HighBP
> # remove HighBP
> m3 <- update(m2, ~ . - HighBP)
>
> anova(m2, m3, test = "LRT")
Analysis of Deviance Table

Model 1: Diabetes ~ HighBP + HighChol + BMI + Age
Model 2: Diabetes ~ HighChol + BMI + Age
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1       115     131.76
2       116     131.98 -1 -0.21671   0.6416
> AIC(m2, m3)
   df      AIC
m2  5 141.7625
m3  4 139.9792
```

3.5 Final Logistic Model and Interpretation

After all the refining process, we can define our final model as:

$$\log\left(\frac{\hat{p}}{1-\hat{p}}\right) = -7.79 + 0.81\,\text{HighChol} + 0.15\text{BMI} + 0.05\text{Age}$$

Still, there are two types of variables in our model, which will be interpreted respectively.

For binary variables, holding other variables constant,

- HighChol, having **high cholesterol,** increases the log odds of diabetes by **0.81.**

For measured variables, holding other variables constant,

- A one-unit increase in **BMI** increases the log odds of diabetes by **0.15.**
- A one-year increase in **age** increases the log odds of diabetes by **0.05.**

```
# Final selected model (after removing Smoker, Fruits, HighBP)
> model_final <- m3   # i.e. Diabetes ~ HighChol + BMI + Age
> summary(model_final)

Call:
glm(formula = Diabetes ~ HighChol + BMI + Age, family =
binomial(link = "logit"),
    data = diabetes_train)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -7.79237    1.84417  -4.225 2.39e-05 ***
HighChol     0.81411    0.43410   1.875 0.060737 .
BMI          0.14929    0.04009   3.724 0.000196 ***
Age          0.05241    0.02359   2.221 0.026335 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 166.06  on 119  degrees of freedom
Residual deviance: 131.98  on 116  degrees of freedom
AIC: 139.98

Number of Fisher Scoring iterations: 4
```

As this is very hard to make sense of, and in order to interpret them into a more sensible fashion, we need to "anti-log" them as the changes in odds ratio for a unit change in variable Xi, while the other variables are held fixed $e(\beta i)$.

The odds of an individual having diabetes (vs not having diabetes) increase by 2.26 when HighChol = 1, all else in the model being fixed. That is, the relative odds

$$\frac{P(Y=1)}{P(Y=0)}$$

are multiplied by e^{0.88} = 2.26, implying an increase of approximately 126%.

- BMI
  The odds of having diabetes increase by 1.16 for a one-unit increase in BMI, holding other variables constant. This corresponds to an increase of approximately 16% in the odds of diabetes.

- Age
  The odds of having diabetes increase by a factor of 1.05 for each additional year of Age, holding all other variables constant, corresponding to an increase of approximately 5% in the odds.

```
> round(exp(coef(model_final)), 4)

(Intercept)    HighChol         BMI         Age

     0.0004      2.2572      1.1610      1.0538
```

3.6 Classification Accuracy and Model Performance

To assess the accuracy, a 0.5 classification threshold is used to translate the final model's predicted probability into binary results; the individual is classified as “diabetes/prediabetes” when $(\hat{p}) \geq 0.5$. There are 10 observations with the FALSE prediction by the final fitted model; the accuracy function suggests that the model accuracy is around 66.67%, which is a normal high prediction.

However, there is some imbalance in the prediction errors: 5 false negatives and 5 false positives. False negatives are those who have diabetes but are incorrectly predicted as having no diabetes, which may cause health issues in further research. Moreover, as there is only a very small sample, only 150, this may cause the risk of overfitting.

```
> confusion_matrix <- table(Y, round(response_pr))

> confusion_matrix

Y    0  1

  0  6  5

  1  5 14

> accuracy <- function(x){

+   sum(diag(x)) / sum(x) * 100

+ }

>

> accuracy(confusion_matrix)

[1] 66.66667
```

3.7 Limitations and brief conclusion

In summary, the model exhibits a reasonable level of explanatory power and prediction accuracy. However, an overemphasis on statistical significance (reading list) may have led to the exclusion of medically important variables such as HighBP. Therefore, more critical analysis is still required.

4. Appendix

Appendix Q1: R Code for Multiple Regression

```
# 2.0) Packages
library(dplyr)
library(ggplot2)
library(GGally)

# 2.0.1) Read data
emp <- read.csv("Employee_Performance.csv", header = TRUE)


# ==========================================================
# 2.2 Data Description and Exploratory Analysis
# ==========================================================
# 1) Quick check
dim(emp)
str(emp)
head(emp)
# 2) Convert categorical variables to factors (so R uses dummy coding)
emp <- emp %>%
  mutate(
    Type = factor(Type, levels = c(0, 1), labels = c("Science", "Business")),
    Level = factor(Level, levels = c(1, 2, 3), labels = c("Degree", "Masters", "Doctorate")),
    Expertise = factor(Expertise, levels = c(1, 2, 3, 4),
                       labels = c("Logistics", "Promotion", "Sales", "Strategy"))
  )

# 3) Descriptive statistics
summary(emp)

# SDs for numeric variables
sapply(emp %>% select(Sales_Perf, Creativity, Mechanical, Abstract, Maths), sd)

# Group counts
table(emp$Type)
table(emp$Level)
table(emp$Expertise)
```

```
# 4) ggpairs on variables
GGally::ggpairs(emp)

# 5) Correlation matrix among numeric variables (supports the multicollinearity comment)
cor(emp %>% select(Sales_Perf, Creativity, Mechanical, Abstract, Maths))

# 6) Boxplots: Sales_Perf by categorical predictors (useful for 2.2 and later interpretation)
ggplot(emp, aes(x = Type, y = Sales_Perf)) + geom_boxplot()
ggplot(emp, aes(x = Level, y = Sales_Perf)) + geom_boxplot()
ggplot(emp, aes(x = Expertise, y = Sales_Perf)) + geom_boxplot()

# confirm reference group in dummy variables
contrasts(emp$Type)
contrasts(emp$Level)
contrasts(emp$Expertise)
# ==============================================================
# 2.3 Initial Model Specification
# ==============================================================
# Initial model
model_initial <- lm(
  Sales_Perf ~ Creativity + Mechanical + Abstract + Maths +
    Type + Level + Expertise,
  data = emp
)
model_initial
summary(model_initial)
# ==============================================================
# 2.4 Model Selection and Refinement
# ==============================================================
# check vif to avoid multicollinearity
library(car)
vif(model_initial)
# Delete Abstract
model_1 <- update(model_initial, . ~ . - Abstract)
summary(model_1)
# Delete Creativity
model_2 <- update(model_1, . ~ . - Creativity)
summary(model_2)
# Delete Expertise
model_3 <- update(model_1, . ~ . - Expertise )
```

```
summary(model_3)
# residual plots
par(mfrow = c(2, 2))
plot(model_2)
par(mfrow = c(1, 1))
# ============================================================
# 2.5 Final Model and Interpretation
# ============================================================
# Using final model to predict
Sales_Perf = 83.0281 + 0.6510 * 18 + 0.2678 * 60 + 10.1580 -0.5668
Sales_Perf
```

Appendix Q2: R Code for Logistic Regression

```
# ============================================================
# 3.0 Load the data
# ============================================================
diabetes <- read.csv("Diabetes.csv")
# ============================================================
# 3.2 Data Description and Exploration
# ============================================================
library(dplyr)
library(ggplot2)
# 1) Function to create count & proportion tables
binary_vars <- c("HighBP", "HighChol", "Smoker", "Fruits")
lapply(binary_vars, function(var) {
  diabetes %>%
    group_by(Diabetes, .data[[var]]) %>%
    summarise(n = n(), .groups = "drop") %>%
    group_by(Diabetes) %>%
    mutate(prop = n / sum(n))
})
# 2) Summary statistics for BMI and Age by diabetes status
diabetes %>%
```

```
  group_by(Diabetes) %>%
  summarise(
    mean_BMI = mean(BMI, na.rm = TRUE),
    sd_BMI   = sd(BMI, na.rm = TRUE),
    min_BMI  = min(BMI, na.rm = TRUE),
    max_BMI  = max(BMI, na.rm = TRUE),

    mean_Age = mean(Age, na.rm = TRUE),
    sd_Age   = sd(Age, na.rm = TRUE),
    min_Age  = min(Age, na.rm = TRUE),
    max_Age  = max(Age, na.rm = TRUE)
  )

# 3) Preparation
library(ggplot2)

# 3.1) Preparation:Recode variables to factors with clear labels ----
diabetes$Diabetes <- factor(
  diabetes$Diabetes,
  levels = c(0, 1),
  labels = c("No diabetes", "Prediabetes / Diabetes")
)

diabetes$Fruits <- factor(
  diabetes$Fruits,
  levels = c(0, 1),
  labels = c("No daily fruit intake", "Daily fruit intake")
)

diabetes$HighBP <- factor(
  diabetes$HighBP,
  levels = c(0, 1),
  labels = c("No high blood pressure", "High blood pressure")
)

diabetes$HighChol <- factor(
  diabetes$HighChol,
  levels = c(0, 1),
  labels = c("No high cholesterol", "High cholesterol")
)
```

```r
diabetes$Smoker <- factor(
  diabetes$Smoker,
  levels = c(0, 1),
  labels = c("Non-smoker", "Smoker")
)

# 3.2)  Common settings
fill_cols <- c("lightblue", "steelblue")  # consistent fill across all proportion plots
bar_border <- "grey30"                    # crisp borders for print/PDF

#  4) plots
# 4.1) Proportion plots:binary predictors vs Diabetes
# Fruits vs Diabetes
ggplot(diabetes, aes(x = Fruits, fill = Diabetes)) +
  geom_bar(position = "fill", colour = bar_border) +
  scale_fill_manual(values = fill_cols) +
  labs(
    x = "Fruit consumption",
    y = "Proportion",
    fill = "Diabetes status"
  ) +
  theme_minimal()

# HighBP vs Diabetes
ggplot(diabetes, aes(x = HighBP, fill = Diabetes)) +
  geom_bar(position = "fill", colour = bar_border) +
  scale_fill_manual(values = fill_cols) +
  labs(
    x = "Blood pressure status",
    y = "Proportion",
    fill = "Diabetes status"
  ) +
  theme_minimal()

# HighChol vs Diabetes
ggplot(diabetes, aes(x = HighChol, fill = Diabetes)) +
  geom_bar(position = "fill", colour = bar_border) +
  scale_fill_manual(values = fill_cols) +
  labs(
```

```
    x = "Cholesterol status",
    y = "Proportion",
    fill = "Diabetes status"
  ) +
  theme_minimal()

# Smoker vs Diabetes
ggplot(diabetes, aes(x = Smoker, fill = Diabetes)) +
  geom_bar(position = "fill", colour = bar_border) +
  scale_fill_manual(values = fill_cols) +
  labs(
    x = "Smoking status",
    y = "Proportion",
    fill = "Diabetes status"
  ) +
  theme_minimal()

# 4.2) Boxplots (measured variables by Diabetes status)

# BMI by Diabetes status
ggplot(diabetes, aes(x = Diabetes, y = BMI)) +
  geom_boxplot(colour = bar_border) +
  labs(
    x = "Diabetes status",
    y = "BMI"
  ) +
  theme_minimal()

# Age by Diabetes status
ggplot(diabetes, aes(x = Diabetes, y = Age)) +
  geom_boxplot(colour = bar_border) +
  labs(
    x = "Diabetes status",
    y = "Age"
  ) +
  theme_minimal()

# ==============================================================
# 3.3. initial logistic regression model
# ==============================================================
```

```r
# 1) Before we start the model fitting procedure we will make test-train split data in the proportion of 80:20, as we will later on assess model accuracy in correct prediction using the test data set
nrow(diabetes)
set.seed(123)
split_idx = sample(nrow(diabetes), 120)

diabetes_train <- diabetes[split_idx, ]
diabetes_test  <- diabetes[-split_idx, ]

# 2) Check the balance
prop.table(table(diabetes_test$Diabetes))
# 3) Initial model
model_initial <- glm(
  Diabetes ~ HighBP + HighChol + BMI + Smoker + Fruits + Age,
  data = diabetes_train,
  family = binomial(link = "logit")
)
# 4) Model summary (coefficients, SEs, z-values, p-values)
summary(model_initial)

# ============================================================
# Section 3.4: Model Selection and Goodness of Fit
# ============================================================

# Overall G test: Full vs Null

# 1) G statistic
G_calc <- model_initial$null.deviance - model_initial$deviance
Gdf    <- model_initial$df.null - model_initial$df.residual

G_calc
Gdf

# 2) Critical value at 5%
G_crit <- qchisq(0.95, df = Gdf)
G_crit

# 3) p-value
p_value_G <- 1 - pchisq(G_calc, df = Gdf)
p_value_G
```

```
# 4) Pseudo R^2
pscl::pR2(model_initial)

# 5) Individual-variable screening using Wald z and p-values
#    (rule of thumb: |z| < 2 and p > 0.05 -> candidate to remove)

full_sum <- summary(model_initial)$coefficients
full_sum

# ==============================================================
# 3.4.1 anova(model, test="Chisq")
# ==============================================================

# 1): fit the full model
# We have fitted the same one in section 2.3


# 2): sequential (Type I) LR tests using deviance table
anova(model_initial, test = "Chisq")

# Model 1: remove Smoker
m1 <- update(model_initial, ~ . - Smoker)

anova(model_initial, m1, test = "LRT")
AIC(model_initial, m1)
# delete smoker
# Model 2: from m1, consider removing fruits(next least significant)
m2 <- update(m1, ~ . - Fruits)

anova(m1, m2, test = "LRT")
AIC(m1, m2)

# remove HighBP
# Model 3: from m2, consider removing HighBP(next least significant)
m3 <- update(m2, ~ . - HighBP)

anova(m2, m3, test = "LRT")
AIC(m2, m3)
# remove HighChol
```

```
# Model 4: from m3, consider removing HighChol
m4 <- update(m3, ~ . - HighChol)

anova(m3, m4, test = "LRT")
AIC(m3, m4)
# ------------------------------------------------------------
# 3.5) Model_final
# ------------------------------------------------------------
# Final selected model (after removing Smoker, Fruits, HighBP)
model_final <- m3   # i.e. Diabetes ~ HighChol + BMI + Age
summary(model_final)

# easier interpretation
round(exp(coef(model_final)), 4)

# ==============================================================
# 3.6 Classification accuracy and model performance
# ==============================================================

# 1) Obtain predicted log-odds and probabilities for the test set
link_pr     <- round(predict(model_final, diabetes_test, type = "link"),     2)
response_pr <- round(predict(model_final, diabetes_test, type = "response"), 2)

# 2) Build a small data frame to inspect predictions
# Convert the factor outcome to 0/1 for convenience
Y <- ifelse(diabetes_test$Diabetes == "Prediabetes / Diabetes", 1, 0)

library(dplyr)

how_well <- data.frame(
  prob      = response_pr,
  actual    = Y,
  predicted = round(response_pr)
) %>%
  mutate(result = (actual == predicted))

how_well

# 3) Confusion matrix: actual (rows) vs predicted (columns)
confusion_matrix <- table(Y, round(response_pr))
```

```
confusion_matrix

# 4) Overall percentage accuracy
accuracy <- function(x){
  sum(diag(x)) / sum(x) * 100
}

accuracy(confusion_matrix)
```